

# MATH 345 Skeleton Notes

## Unit 2: Anatomy of a linear map

### Section 2.3: Subspaces

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### Definition and first examples

The outputs a matrix can “reach” form what is known as a subspace. The directions a matrix “forgets” also form a subspace. The definitions below make this precise.

We are already familiar with geometric spaces living inside bigger ones, like lines lying in the plane, or planes lying in 3-space. We are now going to formalize this notion algebraically. Recall the definition of the set  $\mathbb{R}^n$  in Vector.

#### Definition: Subspaces

A set  $U$  of vectors in  $\mathbb{R}^n$  is called a subspace of  $\mathbb{R}^n$  if it satisfies the following properties:

- The zero vector (written as  $\mathbf{0}$ ) is in  $U$ .
- If  $\mathbf{x}$  is in  $U$  and  $\mathbf{y}$  is in  $U$ , then  $\mathbf{x} + \mathbf{y}$  is in  $U$ . This means  $U$  is *closed under addition*.
- If  $\mathbf{x}$  is in  $U$ , then  $a\mathbf{x}$  is in  $U$  for every scalar  $a$ . This means  $U$  is *closed under scalar multiplication*.

#### Note: Subspaces are closed under linear combinations

Let  $U$  be a subspace. If  $\mathbf{u}, \mathbf{v} \in U$  and  $a, b \in \mathbb{R}$ , closure under scalar multiplication gives  $a\mathbf{u} \in U$  and  $b\mathbf{v} \in U$ . Closure under addition then gives

$$a\mathbf{u} + b\mathbf{v} \in U.$$

More generally, if  $\mathbf{u}_1, \dots, \mathbf{u}_k \in U$  and  $c_1, \dots, c_k \in \mathbb{R}$ , each vector  $c_i\mathbf{u}_i$  lies in  $U$ . Adding these vectors one at a time shows that

$$c_1\mathbf{u}_1 + \dots + c_k\mathbf{u}_k \in U.$$

Thus every finite linear combination of vectors in a subspace remains in that subspace.

### Example

The whole set  $\mathbb{R}^n$  is a subspace of itself.

### Example

The set containing only the zero vector, i.e., the set  $\{\mathbf{0}\}$ , is a subspace of  $\mathbb{R}^n$  (called the trivial subspace or zero subspace).

#### Activity: The zero vector and subspace conditions (U2-LO4)

Verify that  $\{\mathbf{0}\}$  is a subspace of  $\mathbb{R}^n$ .

### Definition: Proper Subspaces

Any subspace of  $\mathbb{R}^n$  other than  $\mathbb{R}^n$  or  $\{0\}$  is a  
of  $\mathbb{R}^n$ .

proper subspace

### Example

Planes and lines through the origin in  $\mathbb{R}^3$  are all subspaces of  $\mathbb{R}^3$ .

### Example: Lines and planes through the origin (U2-LO4)

### Task

Verify that planes through the origin in  $\mathbb{R}^3$  are all proper subspaces of  $\mathbb{R}^3$ .

Let  $R$  be a plane through the origin. Then there exist scalars  $a, b, c$ , not all zero, such that  $R$  is the set of points  $(x, y, z)$  satisfying the equation  $ax + by + cz = 0$ . The vector

$$\mathbf{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

is a normal vector for the plane, and we can write  $R$  as the set of vectors  $\mathbf{v} \in \mathbb{R}^3$  such that  $\mathbf{n} \cdot \mathbf{v} = 0$ .

Let us verify that  $R$  satisfies the conditions to be a subspace:

Contains the zero vector  $\mathbf{0} \in R$  because  $\mathbf{n} \cdot \mathbf{0} = 0$ .

Closure under addition

If  $\mathbf{v} \in R$  and  $\mathbf{w} \in R$  then to verify  $\mathbf{v} + \mathbf{w} \in R$  we must argue that  $\mathbf{n} \cdot (\mathbf{v} + \mathbf{w}) = 0$ . Indeed,

$$\mathbf{n} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{n} \cdot \mathbf{v} + \mathbf{n} \cdot \mathbf{w} = 0.$$

Closure under scalar multiplication

If  $\mathbf{v} \in R$ , and  $a$  is a scalar, to verify  $a\mathbf{v} \in R$  we must argue that  $\mathbf{n} \cdot (a\mathbf{v}) = 0$ . Indeed,

$$\mathbf{n} \cdot (a\mathbf{v}) = a(\mathbf{n} \cdot \mathbf{v}) = a(0) = 0.$$

We have verified the three properties required for  $R$  to be a subspace.

## Task

Verify that lines through the origin in  $\mathbb{R}^3$  are subspaces of  $\mathbb{R}^3$ .

Let  $L$  be a line through the origin in  $\mathbb{R}^3$ . If  $L$  has direction vector  $\mathbf{d}$ , then  $L = \{t\mathbf{d} : t \in \mathbb{R}\}$ , i.e., the line  $L$  consists of all scalar multiples of  $\mathbf{d}$ . Let us verify that  $L$  satisfies the conditions to be a subspace:

Contains the zero vector  $\mathbf{0} \in L$  because  $\mathbf{0} = 0\mathbf{d}$ .

Closure under addition

Suppose  $\mathbf{v} \in L$  and  $\mathbf{w} \in L$ . Then  $\mathbf{v}$  and  $\mathbf{w}$  are both scalar multiples of  $\mathbf{d}$ , i.e.,  $\mathbf{v} = t_1\mathbf{d}$  and  $\mathbf{w} = t_2\mathbf{d}$ . Then

$$\mathbf{v} + \mathbf{w} = t_1 \mathbf{d} + t_2 \mathbf{d} = (t_1 + t_2) \mathbf{d}.$$

Thus  $\mathbf{v} + \mathbf{w}$  is a scalar multiple of  $\mathbf{d}$ , and so  $\mathbf{v} + \mathbf{w} \in L$ .

Closure under scalar multiplication

If  $\mathbf{v} \in L$ , and  $a$  is a scalar, then we may write  $\mathbf{v} = t\mathbf{d}$ , and then

$$a\mathbf{v} = a(t\mathbf{d}) = (at)\mathbf{d}.$$

So  $a\mathbf{v}$  is a scalar multiple of  $\mathbf{d}$ , and thus  $a\mathbf{v} \in L$ . We have verified the three properties required for  $L$  to be a subspace.

### Note

Lines and planes which do not pass through the origin in  $\mathbb{R}^3$  are *not* subspaces of  $\mathbb{R}^3$ . Nonetheless, if  $\Pi$  is a plane, and we consider the set of displacement vectors

$$\Delta\Pi = \{\mathbf{v}_1 - \mathbf{v}_0 : \mathbf{v}_0, \mathbf{v}_1 \in \Pi\},$$

then  $\Delta\Pi$  forms a plane passing through the origin, and so is a subspace of  $\mathbb{R}^3$ . A similar result holds for sets of displacement vectors given by a line not passing through the origin.

### Example: A product condition on coordinates (U2-LO4)

Show that the set

$$U = \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : x_1 x_2 = 0 \right\}$$

is *not* a subspace of  $\mathbb{R}^2$ .

It suffices to show *one* of the conditions required to be a subspace does not hold. *U does* contain the zero vector, and is closed under scalar multiplication. But *U is not* closed under addition. Define

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then  $\mathbf{x} \in U$  and  $\mathbf{y} \in U$ . But

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

is not in  $U$ , since  $(1)(1) \neq 0$ . Since  $U$  fails to be closed under addition, it is not a subspace of  $\mathbb{R}^2$ .

**Activity: A set described by two parameters (U2-LO4)**

Let  $W$  be the set of all vectors in  $\mathbb{R}^3$  of the form

$$\begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix},$$

where  $r, s \in \mathbb{R}$ . Is  $W$  a proper subspace of  $\mathbb{R}^3$ ?





## Note

To show a set *is* a subspace, you tend to need to parameterize all elements of the set in order to show they all satisfy some property (closure under addition or scalar multiplication), and so your argument should use variables. To show a set *is not* a subspace, you need to find one or more counterexamples that show that some property does not hold for all elements; a single explicit counterexample usually suffices.

## Null space and image space

The following important examples are representative of two canonical ways of describing a subspace.

### Definition: Null Space and Image Space

Let  $A$  be an  $m \times n$  matrix. The null space of  $A$ , denoted  $\text{null}(A)$ , is the set of vectors  $\mathbf{x}$  in  $\mathbb{R}^n$  which are solutions to the equation  $A\mathbf{x} = \mathbf{0}$ , i.e.,

$$\text{null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}.$$

The **image space** of  $A$ , denoted  $\text{im}(A)$ , is the set of vectors  $\mathbf{y}$  in  $\mathbb{R}^m$  such that  $A\mathbf{x} = \mathbf{y}$  has a solution, i.e.,

$$\text{im}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}.$$

### Note

The null space lives in the input space. It records input directions that  $A$  sends to  $\mathbf{0}$ .

The image space lives in the output space. It records outputs that  $A$  can reach.

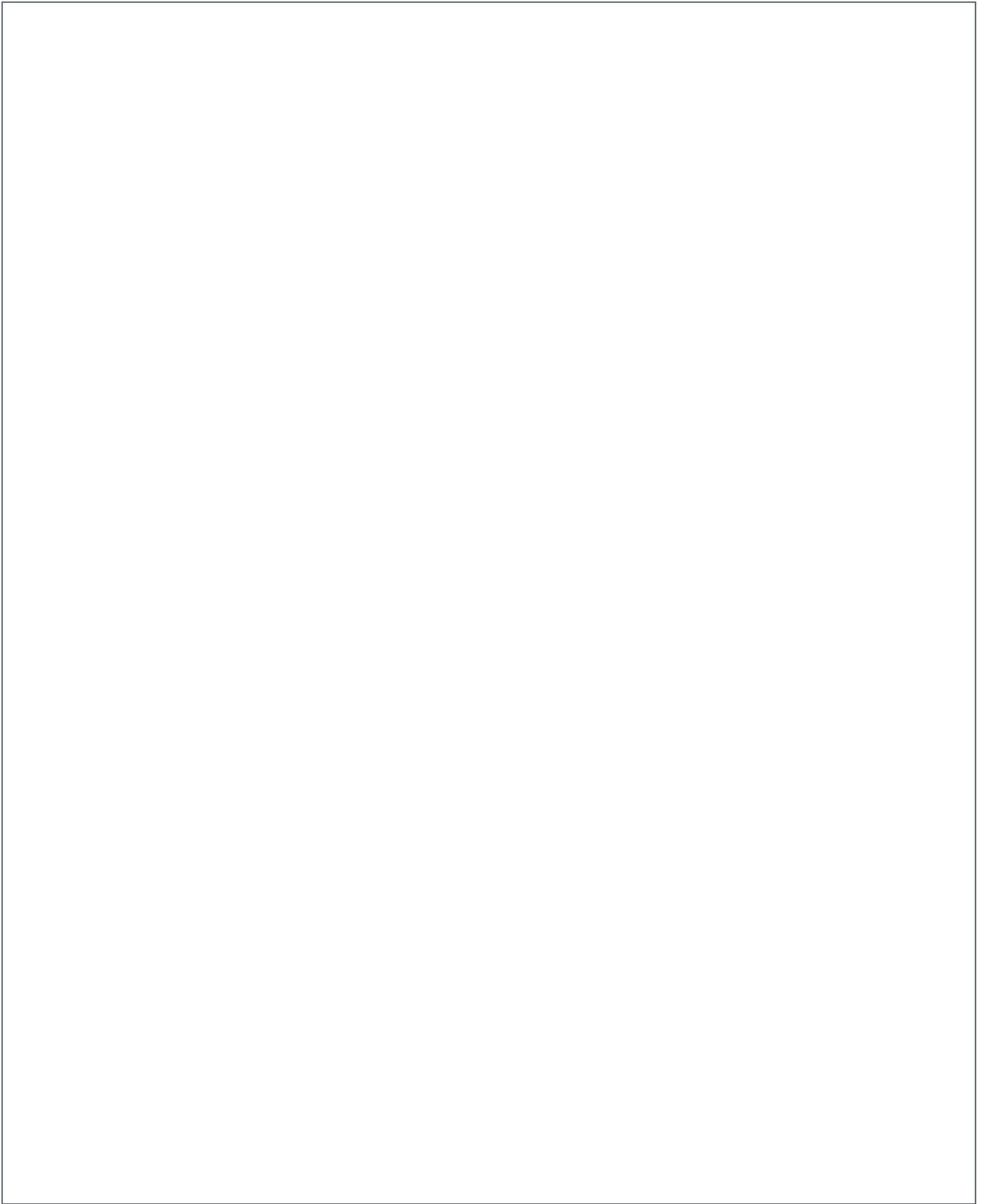
- **Object:**  $\text{null}(A)$   
**Lives in:**  $\mathbb{R}^n$   
**How to read it:** input directions  $A$  sends to  $\mathbf{0}$
- **Object:**  $\text{im}(A)$   
**Lives in:**  $\mathbb{R}^m$   
**How to read it:** outputs  $A$  can reach

### Activity: Revisiting the height map with subspace language (U2-LO3)

Return to  $A$  map that forgets height (U2-LO1, U2-LO3). Let

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Describe the output subspace and the subspace of input directions that  $P$  sends to  $\mathbf{0}$ .



## Theorem

For any  $m \times n$  matrix, the null space of  $A$  is a subspace of  $\mathbb{R}^n$ , and the image space of  $A$  is a subspace of  $\mathbb{R}^m$ .

### Why is this true?

We start by showing the null space of  $A$  is a subspace. Recall the properties discussed in Properties of matrix-vector products:

Contains the zero vector  $\mathbf{0} \in \text{null}(A)$  because  $A\mathbf{0} = \mathbf{0}$ .

Closure under addition

Suppose  $\mathbf{v} \in \text{null}(A)$  and  $\mathbf{w} \in \text{null}(A)$ . Then  $A\mathbf{v} = \mathbf{0}$  and  $A\mathbf{w} = \mathbf{0}$ . So

$$A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w} = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

Thus  $\mathbf{v} + \mathbf{w} \in \text{null}(A)$ .

Closure under scalar multiplication

If  $\mathbf{v} \in \text{null}(A)$ , and  $a$  is a scalar, then

$$A(a\mathbf{v}) = a(A\mathbf{v}) = a\mathbf{0} = \mathbf{0}.$$

Thus  $a\mathbf{v} \in \text{null}(A)$ .

So  $\text{null}(A)$  is a subspace of  $\mathbb{R}^n$ .

Next, we show that  $\text{im}(A)$  is a subspace.

Contains the zero vector  $\mathbf{0} \in \text{im}(A)$  because  $A\mathbf{0} = \mathbf{0}$ .

Closure under addition

Suppose  $\mathbf{v} \in \text{im}(A)$  and  $\mathbf{w} \in \text{im}(A)$ . Then the equations  $A\mathbf{x} = \mathbf{v}$  and  $A\mathbf{x} = \mathbf{w}$  both have solutions, i.e., there exists  $\mathbf{x}_1$  and  $\mathbf{x}_2$  such that  $A\mathbf{x}_1 = \mathbf{v}$  and  $A\mathbf{x}_2 = \mathbf{w}$ . Then

$$A(\mathbf{x}_1 + \mathbf{x}_2) = A\mathbf{x}_1 + A\mathbf{x}_2 = \mathbf{v} + \mathbf{w},$$

so the equation  $A\mathbf{x} = \mathbf{v} + \mathbf{w}$  has a solution, and so  $\mathbf{v} + \mathbf{w} \in \text{im}(A)$ .

Closure under scalar multiplication

If  $\mathbf{v} \in \text{im}(A)$ , and  $a$  is a scalar, then we may write  $\mathbf{v} = A\mathbf{x}_1$  for some  $\mathbf{x}_1$ .

But then

$$A(ax_1) = a(Ax_1) = a\mathbf{v},$$

so the equation  $A\mathbf{x} = a\mathbf{v}$  has a solution, so that  $a\mathbf{v} \in \text{im}(A)$ . Thus  $\text{im}(A)$  is a subspace of  $\mathbb{R}^m$ .

### Warning

The solution set of  $A\mathbf{x} = \mathbf{0}$  is a subspace. The solution set of  $A\mathbf{x} = \mathbf{b}$  for  $\mathbf{b} \neq \mathbf{0}$  is usually not a subspace; when it is nonempty, it is a shifted copy of  $\text{null}(A)$ .

From here on, homogeneous equations play a special role. To solve  $A\mathbf{x} = \mathbf{0}$  by row reduction, we could write the augmented matrix  $[A \mid \mathbf{0}]$ . Since the right-hand side stays zero throughout the row operations, the coefficient matrix  $A$  carries the essential information.

## Spanning Sets

Even though subspaces of  $\mathbb{R}^n$  can have infinitely many vectors, it is possible to represent them with a finite amount of data.

### Activity: Writing a parametric set as a span (U2-LO4)

Recall from A set described by two parameters (U2-LO4) that

$$W = \left\{ \begin{bmatrix} r \\ s \\ r + 2s \end{bmatrix} : r, s \in \mathbb{R} \right\}.$$

Show that there are two vectors (describe them explicitly), so that every vector in  $W$  can be written as a linear combination of those two vectors.

### Definition: The Span of a Set of Vectors

The set of all linear combinations of a set of vectors  $\mathbf{x}_1, \dots, \mathbf{x}_k$  is called the **span** of the vectors  $\mathbf{x}_1, \dots, \mathbf{x}_k$ , and is denoted

$$\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\} = \{t_1\mathbf{x}_1 + \dots + t_k\mathbf{x}_k : t_1, \dots, t_k \in \mathbb{R}\}.$$

If  $V = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ , we say that  $V$  is **spanned** by the vectors  $\mathbf{x}_1, \dots, \mathbf{x}_k$ , and that the vectors  $\mathbf{x}_1, \dots, \mathbf{x}_k$  **span** the space  $V$ .

### Example

Recall the standard basis  $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$  of  $\mathbb{R}^n$  from Standard basis. Note that

if

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

then

$$\mathbf{x} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + \cdots + x_n \mathbf{e}_n.$$

This means that  $\mathbb{R}^n = \text{span}\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ .

For an image space, spanning vectors come from the columns of the matrix. If

$$A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]$$

and

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

then

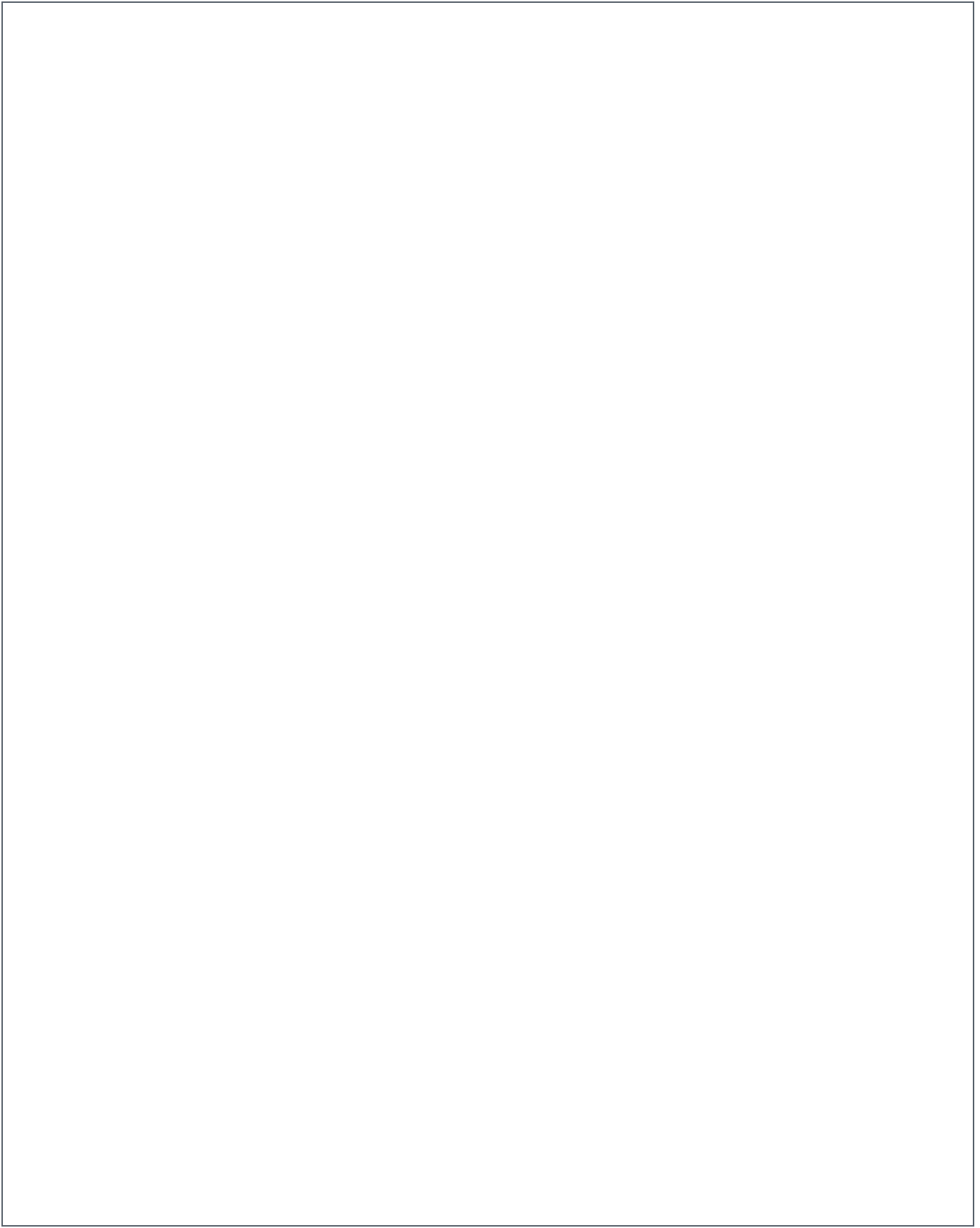
$$A\mathbf{x} = x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n.$$

Thus every output  $A\mathbf{x}$  is a linear combination of the columns of  $A$ .

**Activity: A spanning set for an image space (U2-LO3, U2-LO4)**

Find a spanning set for the image space of the matrix

$$A = \begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & -1 & -3 & 2 \\ 2 & 1 & 7 & -4 \end{bmatrix}.$$



**Activity: A spanning set for a null space (U2-LO3, U2-LO4)**

Find a spanning set for the null space of the matrix

$$A = \begin{bmatrix} 1 & -2 & -4 & 3 \\ 0 & -1 & -3 & 2 \\ 2 & 1 & 7 & -4 \end{bmatrix}.$$

Practice this activity in [The Elimination Scratchpad](#)



To summarize: basic solutions of  $A\mathbf{x} = \mathbf{0}$  form a spanning set for  $\text{null}(A)$ , and the columns of  $A$  form a spanning set for  $\text{im}(A)$ .

### Theorem

Let  $U = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$  be a set in  $\mathbb{R}^n$ . Then

- $U$  is a *subspace* of  $\mathbb{R}^n$  containing each of the vectors  $\mathbf{x}_1, \dots, \mathbf{x}_k$ .
- If  $W$  is a subspace of  $\mathbb{R}^n$  and each of the vectors  $\mathbf{x}_1, \dots, \mathbf{x}_k$  is in  $W$ , then  $U$  is a *subset* of  $W$ , i.e.,  $U \subseteq W$ .

### Why is this true?

Define an  $n \times k$  matrix  $A$ , with the vectors  $\mathbf{x}_1, \dots, \mathbf{x}_k$  as columns. Then  $U = \text{im}(A)$ , and is thus a subspace of  $\mathbb{R}^n$ . Since  $\mathbf{x}_i = A\mathbf{e}_i$  for  $1 \leq i \leq k$ , the set  $U$  contains each of the vectors  $\mathbf{x}_i$ .

Conversely, suppose  $W$  is a subspace of  $\mathbb{R}^n$  and  $\mathbf{x}_1, \dots, \mathbf{x}_k \in W$ . Since

$W$  is closed under addition and scalar multiplication, all of the linear combinations of  $\mathbf{x}_1, \dots, \mathbf{x}_k$  are elements of  $W$ , and so  $U \subseteq W$ .

**Activity: Testing membership in a span (U2-LO4)**

In  $\mathbb{R}^3$ , consider the two vectors

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}.$$

Determine if the vector

$$\mathbf{w} = \begin{bmatrix} 1 \\ 5 \\ -7 \end{bmatrix}$$

belongs to  $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ .





### Activity: Testing membership in another span (U2-LO4)

In  $\mathbb{R}^3$ , consider the two vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

Determine if the vector

$$\mathbf{w} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

belongs to  $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$ .

The next section studies when a spanning set has redundant vectors and how to remove them.